

Web Layer



«Controller»
WeatherController

• getWeather(lat, lng): ResponseEntity<WeatherResponse>

Business Logic



«Service»
WeatherService

• getWeather(lat, lng): WeatherResponse
■ roundCoord(value): double
■ callForecastApi(lat, lng): CurrentWeather
■ callMarineApi(lat, lng): CurrentMarine
■ buildWeatherResponse(atm, mar): WeatherResponse
■ evaluateDivingCondition(wr): String
■ saveToCache(rLat, rLng, wr)
■ convertFromCache(cw): WeatherResponse

HTTP GET

HTTP GET

APIs Externas



«External»
ForecastAPI

<https://api.open-meteo.com/v1/forecast>

temperature, humidity
windSpeed, windDirection
precipitation, visibility
weatherCode (WMO)



«External»
MarineAPI

<https://marine-api.open-meteo.com/v1/marine>

waveHeight, wavePeriod
seaSurfaceTemperature
oceanCurrentVelocity
swellWaveHeight

Persistence



«Repository»
CachedWeatherRepository

• findFirstByRoundedLatAndRoundedLng
AndCreatedAtAfter(lat, lng, cutoff)

Entities



«Entity»
CachedWeather

□ id: Long
□ roundedLat: Double
□ roundedLng: Double
□ createdAt: Instant
□ temperature: Double
□ windSpeed: Double
□ visibility: Double
□ weatherCode: Integer
□ waveHeight: Double
□ oceanCurrentVelocity: Double
□ seaSurfaceTemperature: Double
□ divingCondition: String

Icono	Visibilidad (Modificador)
●	Público (+): Visible para todos
■	Privado (-): Solo visible dentro de la clase
□	Protegido / Package (# / ~): Acceso restringido
Color	Capa Arquitectónica
■	Controller: Entrada de la API
■	Service: Lógica de negocio
■	Repository: Acceso a Datos
■	Entity: Modelos de DB
■	External: API Externa